

Soft-Motif MoS-NMF

1 Purpose

This note describes the target mixture-of-submanifolds NMF model in which the submanifold motifs are random. It extends the current hard-motif implementation by replacing the hard motif assignment

$$a_{sd} \in [K]$$

with a random factor indicator

$$\gamma_{sd} \in [K],$$

and by learning a posterior distribution over these indicators.

The goal is to quantify uncertainty in submanifold-level sparsity patterns. This is especially important when several fitted factor rows are highly correlated or when the number of fitted factors is overspecified.

2 Model

We observe a count matrix $Y \in \mathbb{N}^{N \times M}$. Let $F \in \mathbb{R}_+^{K \times M}$ be a row-normalized factor matrix:

$$F_{km} \geq 0, \quad \sum_{m=1}^M F_{km} = 1, \quad k \in [K].$$

There are S submanifolds and each submanifold has nominal dimension D . For submanifold s and slot d , the factor identity is random:

$$\gamma_{sd} \sim \text{Cat}(\rho_{sd1}, \dots, \rho_{sdK}).$$

The simplest prior is uniform, $\rho_{sdk} = 1/K$. More generally, ρ_{sd} can be learned or given a hierarchical prior.

For each observation i :

$$Z_i \sim \text{Cat}(\pi_1, \dots, \pi_S), \tag{1}$$

$$\beta_{id} \mid Z_i = s \sim \text{Gam}(\alpha_{sd}^0, \lambda_{sd}^0), \quad d \in [D], \tag{2}$$

$$Y_{im} \mid Z_i = s, \gamma, \beta_i, F \sim \text{Poi} \left(\sum_{d=1}^D \beta_{id} F_{\gamma_{sd}, m} \right), \quad m \in [M]. \tag{3}$$

Equivalently, the loading vector for observation i is

$$L_i = \sum_{d=1}^D e_{\gamma_{Z_i d}} \beta_{id},$$

where e_k is the k th coordinate vector in $\mathbb{R}^K$.

Remark 1 (Relationship to the hard-motif model). The hard-motif model is recovered when each posterior or prior over γ_{sd} is a point mass. The hard motif a_{sd} corresponds to $q(\gamma_{sd} = a_{sd}) = 1$.

3 Variational family

We use a mean-field approximation that keeps the submanifold motif indicators global:

$$q(Z, \gamma, \beta) = \prod_{i=1}^N q(Z_i) \prod_{s=1}^S \prod_{d=1}^D q(\gamma_{sd}) \prod_{i=1}^N \prod_{s=1}^S \prod_{d=1}^D q(\beta_{id} \mid Z_i = s).$$

Write

$$q(Z_i = s) = \tilde{z}_{is}, \quad q(\gamma_{sd} = k) = \tilde{\gamma}_{sdk}, \quad q(\beta_{id} \mid Z_i = s) = \text{Gam}(\alpha_{isd}, \lambda_{isd}).$$

We introduce auxiliary variables

$$\xi_{isd m} \geq 0, \quad \sum_{d=1}^D \xi_{isd m} = 1,$$

for each observation i , submanifold s , and coordinate m . These allocate the count Y_{im} across the D slots of candidate submanifold s .

4 Lower bound

For fixed i, s, m , Jensen's inequality gives

$$\log \left(\sum_{d=1}^D \beta_{id} F_{\gamma_{sd}, m} \right) \geq \sum_{d=1}^D \xi_{isd m} \log \left(\frac{\beta_{id} F_{\gamma_{sd}, m}}{\xi_{isd m}} \right).$$

Ignoring constants $-\log(Y_{im}!)$, the variational lower bound is

$$\begin{aligned} \mathcal{L} = & \sum_{i,s} \tilde{z}_{is} \sum_{d,m} \xi_{isd m} Y_{im} \{ \mathbb{E}_q[\log \beta_{id} \mid Z_i = s] + \mathbb{E}_q[\log F_{\gamma_{sd}, m}] - \log \xi_{isd m} \} \\ & - \sum_{i,s} \tilde{z}_{is} \sum_{d,m} \mathbb{E}_q[\beta_{id} \mid Z_i = s] \mathbb{E}_q[F_{\gamma_{sd}, m}] \\ & - \sum_{i,s} \tilde{z}_{is} \sum_d \text{KL}\{q(\beta_{id} \mid Z_i = s) \parallel p(\beta_{id} \mid Z_i = s)\} \\ & - \sum_{s,d} \text{KL}\{q(\gamma_{sd}) \parallel p(\gamma_{sd})\} + \sum_{i,s} \tilde{z}_{is} (\log \pi_s - \log \tilde{z}_{is}). \end{aligned} \tag{4}$$

The expectations over γ are

$$\mathbb{E}_q[\log F_{\gamma_{sd}, m}] = \sum_{k=1}^K \tilde{\gamma}_{sdk} \log F_{km}, \tag{5}$$

$$\mathbb{E}_q[F_{\gamma_{sd}, m}] = \sum_{k=1}^K \tilde{\gamma}_{sdk} F_{km}. \tag{6}$$

5 Coordinate updates

5.1 Auxiliary allocation

For each i, s, m, d ,

$$\xi_{isdm} \leftarrow \frac{\exp \{ \mathbb{E}_q[\log \beta_{id} \mid Z_i = s] + \sum_k \tilde{\gamma}_{sdk} \log F_{km} \}}{\sum_{d'=1}^D \exp \{ \mathbb{E}_q[\log \beta_{id'} \mid Z_i = s] + \sum_k \tilde{\gamma}_{sd'k} \log F_{km} \}}.$$

5.2 Gamma effect-size posterior

For each i, s, d ,

$$\alpha_{isd} \leftarrow \alpha_{sd}^0 + \sum_{m=1}^M \xi_{isdm} Y_{im}, \quad (7)$$

$$\lambda_{isd} \leftarrow \lambda_{sd}^0 + \sum_{m=1}^M \sum_{k=1}^K \tilde{\gamma}_{sdk} F_{km}. \quad (8)$$

With row-normalized F , the second term is 1 for any valid probability vector $\tilde{\gamma}_{sd}$, but keeping the expression explicit is useful for checking scale conventions.

5.3 Submanifold motif posterior

For each submanifold s , slot d , and candidate factor k , collect the terms in the ELBO that depend on $\gamma_{sd} = k$. Dropping constants in k gives

$$\ell_{sdk} = \log \rho_{sdk} + \sum_{i=1}^N \tilde{z}_{is} \left[\sum_{m=1}^M \xi_{isdm} Y_{im} \log F_{km} - \frac{\alpha_{isd}}{\lambda_{isd}} \sum_{m=1}^M F_{km} \right]. \quad (9)$$

Then

$$\tilde{\gamma}_{sdk} \leftarrow \frac{\exp(\ell_{sdk})}{\sum_{\ell=1}^K \exp(\ell_{sd\ell})}.$$

If F is row-normalized, the term $\sum_m F_{km}$ is constant in k and drops out. In that case the update is driven by responsibility-weighted allocated counts matched against $\log F_{k\cdot}$, plus the prior ρ_{sd} .

Remark 2 (Posterior uncertainty in motifs). The vector $\tilde{\gamma}_{sd} = (\tilde{\gamma}_{sd1}, \dots, \tilde{\gamma}_{sdK})$ is the submanifold-level analog of a SuSiE PIP vector. Diffuse $\tilde{\gamma}_{sd}$ means that slot d of submanifold s is uncertain across several factor rows. This is the object that should quantify uncertainty when overfitted or highly correlated factor rows compete to explain the same sparsity pattern.

5.4 Mixture responsibilities

Define the component lower bound contribution for observation i under submanifold s :

$$\begin{aligned} \mathcal{L}_{is} = & \sum_{d,m} \xi_{isdm} Y_{im} \left\{ \psi(\alpha_{isd}) - \log \lambda_{isd} + \sum_k \tilde{\gamma}_{sdk} \log F_{km} - \log \xi_{isdm} \right\} \\ & - \sum_{d,m} \frac{\alpha_{isd}}{\lambda_{isd}} \sum_k \tilde{\gamma}_{sdk} F_{km} - \sum_d \text{KL}\{\text{Gam}(\alpha_{isd}, \lambda_{isd}) \parallel \text{Gam}(\alpha_{sd}^0, \lambda_{sd}^0)\}. \end{aligned} \quad (10)$$

Then

$$\tilde{z}_{is} \leftarrow \frac{\pi_s \exp(\mathcal{L}_{is})}{\sum_{s'=1}^S \pi_{s'} \exp(\mathcal{L}_{is'})}.$$

The mixture weights are updated by

$$\pi_s \leftarrow \frac{1}{N} \sum_{i=1}^N \tilde{z}_{is}.$$

5.5 Empirical-Bayes prior update

For each submanifold s and slot d , compute responsibility-weighted moments

$$\bar{\beta}_{sd} = \frac{\sum_i \tilde{z}_{is} \alpha_{isd} / \lambda_{isd}}{\sum_i \tilde{z}_{is}}, \quad (11)$$

$$\overline{\log \beta}_{sd} = \frac{\sum_i \tilde{z}_{is} \{\psi(\alpha_{isd}) - \log \lambda_{isd}\}}{\sum_i \tilde{z}_{is}}. \quad (12)$$

The Gamma shape solves

$$\log \alpha_{sd}^0 - \psi(\alpha_{sd}^0) = \log \bar{\beta}_{sd} - \overline{\log \beta}_{sd},$$

and

$$\lambda_{sd}^0 \leftarrow \alpha_{sd}^0 / \bar{\beta}_{sd}.$$

5.6 Factor update

The terms involving F_{km} are weighted by the posterior probability that a submanifold slot uses factor k . The expected allocated count is

$$A_{km} = \sum_{i=1}^N \sum_{s=1}^S \sum_{d=1}^D \tilde{z}_{is} \tilde{\gamma}_{sdk} \xi_{isd m} Y_{im}.$$

With row-normalized F , the update is

$$F_{km}^{\text{new}} \leftarrow \frac{A_{km} + \epsilon}{\sum_{m'=1}^M (A_{km'} + \epsilon)}.$$

A damped update can be used:

$$F_{k\cdot} \leftarrow (1 - \eta_t) F_{k\cdot} + \eta_t F_{k\cdot}^{\text{new}}, \quad F_{k\cdot} \leftarrow F_{k\cdot} / \sum_m F_{km}.$$

6 Algorithm

Algorithm 1 Soft-motif MoS-NMF

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1: Initialize  $F$ , for example by MF-Poisson-SuSiE with learned  $F$ .
2: Initialize  $\tilde{z}_{is}$  by clustering posterior loading scores or by random assignment.
3: Initialize  $\tilde{\gamma}_{sdk}$  from cluster-level loading patterns, from hard motifs, or randomly.
4: Initialize  $\alpha_{isd}$ ,  $\lambda_{isd}$ , and  $\xi_{isd m} = 1/D$ .
5: for  $t = 1, \dots, T$  do
6:   for  $s = 1, \dots, S$  do
7:     for  $d = 1, \dots, D$  do
8:       Update  $\xi_{isd m}$  for all  $i, m$  using current  $\tilde{\gamma}$ ,  $\alpha$ , and  $\lambda$ .
9:       Update  $\alpha_{isd}$  and  $\lambda_{isd}$  for all  $i$ .
10:      Update motif posterior  $\tilde{\gamma}_{sdk}$  for all  $k$ .
11:      Optionally update  $(\alpha_{sd}^0, \lambda_{sd}^0)$ .
12:    end for
13:  end for
14:  Update component lower bounds  $\mathcal{L}_{is}$ .
15:  Update mixture responsibilities  $\tilde{z}_{is}$  and mixture weights  $\pi_s$ .
16:  Optionally update  $F$  using expected counts  $A_{km} = \sum_{i,s,d} \tilde{z}_{is} \tilde{\gamma}_{sdk} \xi_{isd m} Y_{im}$ .
17:  Evaluate or monitor the ELBO in Eq. (??).
18: end for

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7 Relationship to current implementation

The current implemented hard-motif algorithm uses a matrix $a_{sd} \in [K]$ and responsibilities ω_{is} . It corresponds to the restricted variational family

$$\tilde{\gamma}_{sdk} = \mathbf{1}\{k = a_{sd}\}.$$

The soft-motif model replaces this point mass with a full categorical posterior for each submanifold slot. Under this replacement:

- the hard count update for F becomes an expected count update weighted by $\tilde{\gamma}_{sdk}$;
- hard motif changes become categorical posterior updates;
- uncertainty in overfitted or correlated factor rows is represented by diffuse $\tilde{\gamma}_{sd}$.

8 Open implementation choices

Remark 3 (Observation-specific versus submanifold-specific effects). The model above keeps γ_{sd} global to submanifold s while β_{id} remains observation-specific. This is the intended submanifold-level sparsity pattern model. An alternative would give each observation its own γ_{isd} while tying it through a submanifold-level prior, but that would be closer to a hierarchical MF-Poisson-SuSiE model than to a motif model.

Remark 4 (Avoiding duplicated motif slots). Independent categorical posteriors over γ_{sd} allow two slots in the same submanifold to choose the same factor. This is useful when the effective dimension is smaller than D , but it can also duplicate active slots. Possible regularizations include symmetry breaking, repulsive priors, or a without-replacement motif prior.

Remark 5 (Monotonicity). If all updates are derived from the same lower bound and are performed exactly, the coordinate ascent should be monotone. In practice, damping, initialization heuristics, symmetry breaking, or approximate updates may make the monitored objective non-monotone.